New trading relationships can change the dominant vehicle.

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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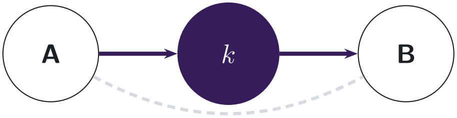
A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider. Project Rialto uses a vehicle when direct exchange is unavailable.



Currency k bridges an illiquid pair.

vehicle currency

Winning the middle concentrates trading and liquidity in the vehicle.

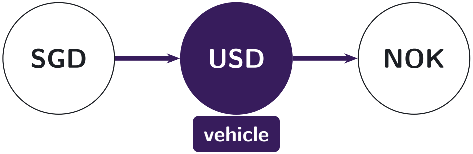
Note: Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. We study the intermediary asset observed inside DEX pool routes.

Pool routes reveal the vehicle currency

SINGAPORE PAYMENT ANALOGY

A Singapore-dollar payment to Norway may be converted through US dollars.

Conventional turnover data show two legs, not the customer's connected route.



Pool logs reveal the route.

DeFi is on-chain finance executed by smart contracts; decentralised exchanges (DEXs) trade tokens through liquidity pools. **Stablecoins** are tokens designed to track a reference currency, usually the US dollar.

Note: The currency labels illustrate a possible indirect conversion, not an observed customer route. Conventional FX records report turnover in each currency pair but generally do not link the two legs to one instruction. The DEX setting preserves the transaction-level pool route, so the intermediary asset is observed directly.

The route panel links pool-level swaps

475 million
pool-level swaps

2,798
calendar dates

9
DEX deployments

November 2018–June 2026

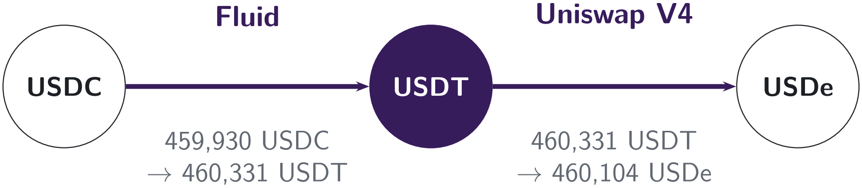
Uniswap v1, v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid.

Note: The exchange families were selected for Ethereum volume. They account for 87.5% of total Ethereum DEX volume in DeFiLlama from 2020 through 2026 H1. The nine deployments span major constant-product, concentrated-liquidity, stable-swap, weighted-pool, singleton, and integrated-liquidity designs. Other Ethereum venues remain outside the panel. The exact exchange registry maps every observed Uniswap v1 exchange to its token.

Inside the pools, USDT links USDC to USDe

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION

User instruction: PYUSD $\rightarrow$ USDe	$\Longrightarrow$	Pool route: USDC $\rightarrow$ USDT $\rightarrow$ USDe



vehicle

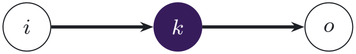
The ordered endpoint pair, or pair, is USDC $\rightarrow$ USDe. Its legs are USDC $\rightarrow$ USDT and USDT $\rightarrow$ USDe; USDT is the vehicle.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026. The pool component begins at USDC. It does not reveal whether PYUSD became USDC in an unobserved pool or through the executor's inventory.

The full route and the one-vehicle choice answer different questions

MEASUREMENT

- Every complete route contributes each currency used in an intermediary position.
- Exact two-leg routes isolate one vehicle choice.

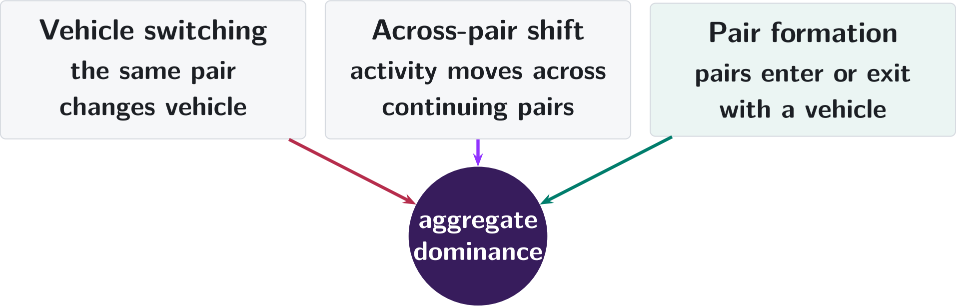


All route lengths
Presence on a route and share of all intermediary positions

One vehicle per route
Exact two-leg routes support the decomposition

Note: Intermediary-position shares sum to 100%. Route-participation shares can exceed 100% because long routes use several currencies. Exact two-leg routes isolate one vehicle choice; longer routes can use both families.

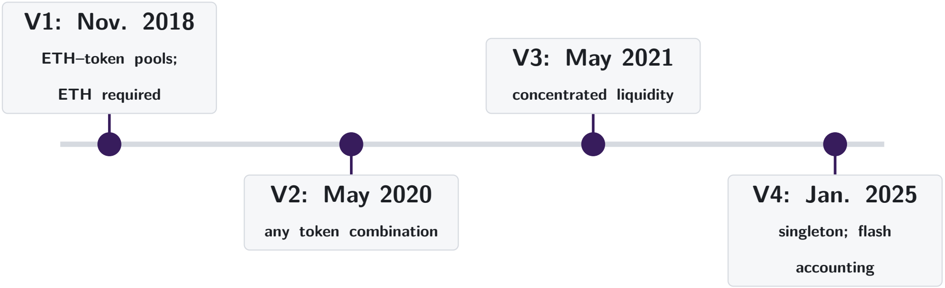
Aggregate dominance can change in three ways



Note: The decomposition is exact. The next step asks why a pair keeps its first vehicle and when a competing two-leg bridge becomes deep enough to attract routes.

Protocol design changes the route set in steps

UNISWAP ON ETHEREUM



V2 opens vehicle choice; later versions change how liquidity is supplied.

Note: First-swap dates. The next slide isolates V1 to V2: compulsory ETH routing ends while the inherited market structure remains.

Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every pool is an ETH–token pool, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: POOL FORMATION

95.5%

of single-leg V2 trades use a WETH pool in 2026

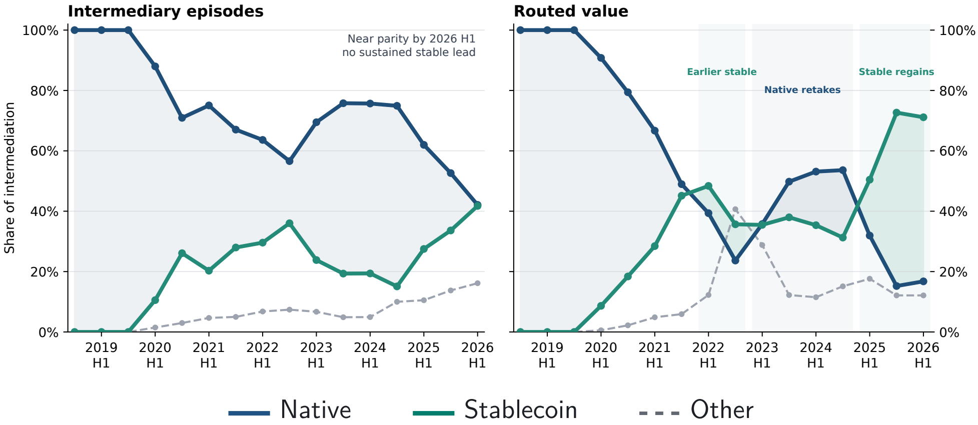
97.9%

of token combinations first traded in 2026 include WETH

Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports legs and newly traded token combinations on that venue. Early V2 inherited liquidity, users, and routing conventions from V1, so part of the initial persistence is mechanical; later pool formation shows how long that structure lasts.

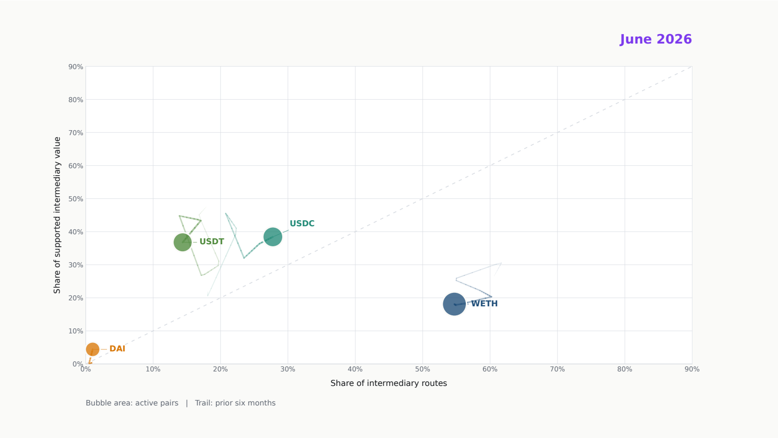
Stablecoins regain the routed-value lead by 2026 H1

ALL ROUTE LENGTHS



Note: Each complete route contributes once for every intermediary position, including routes with more than two legs. Every point covers six calendar months; the final point is 2026 H1. Routed value requires source, intermediary, and destination values to agree within 20%.

Vehicle leadership turns over through time



Click the film to play in Adobe Acrobat

Note: The film follows WETH, USDC, USDT, and DAI in route participation, supported value, and pair breadth. Each frame retains six months of history, so no single frame contains the full transition. The poster is the PDF fallback.

Pair formation supplies most of the rotation

EXACT DECOMPOSITION IN PERCENTAGE POINTS (PP): 2024 H1 TO 2026 H1

Same continuing pair	Activity shifts across pairs	Pairs in one period
-0.1 **pp**	$+8.4$ **pp**	$+17.8$ **pp**

Stablecoin share

2024 H1: **16.9%** → 2026 H1: **42.4%**

The aggregate rises $+25.5$ pp; net switching inside continuing pairs is -0.1 pp.

Note: Exact two-leg native-or-stable vehicle routes on matched January–June dates. A further -0.5 pp comes from the changing weight of continuing versus period-specific pairs. Components use unrounded values and sum to the aggregate change.

The first vehicle is sticky

PREDICTIVE PERSISTENCE AFTER PAIR ENTRY

INITIAL STABLECOIN SHARE

+1 pp

at pair entry

$\Longrightarrow$

NEXT 30 DAYS

$+0.86$ **pp**

stablecoin share

$\Longrightarrow$

NEXT 120 DAYS

$+0.74$ **pp**

stablecoin share

The first vehicle remains a focal point for the new trading relationship.

Note: 599,870 non-WETH pairs observed by route scope. Regressions control for entry size, cohort year, endpoint class, direct-route share, and route complexity. Standard errors are clustered by entry date. "Sticky" means predictive persistence in observed route use; it does not identify trader inertia.

A shallow stablecoin bridge attracts little route flow

FIRST 30 DAYS AFTER PERSISTENT STABLECOIN SUPPORT

Below $0.1\times$ WETH depth	→	At least WETH depth	→	At least $2\times$ WETH depth
2.4%		**53.0%**		**69.9%**

Vehicle competition turns on relative depth.

Note: Depth is prior-calendar deposited capital on the weaker leg. The route-share comparison uses 11,327 pair-days across 855 bridge events. Support and depth refer to the same stablecoin set. Capital and route demand are jointly determined.

Usable depth turns support into route use

PROBABILITY OF FIRST STABLECOIN USE WITHIN 30 DAYS

BELOW $0.1\times$ WETH DEPTH

42.6%

$\rightarrow$

AT LEAST $0.1\times$ WETH DEPTH

84.1%

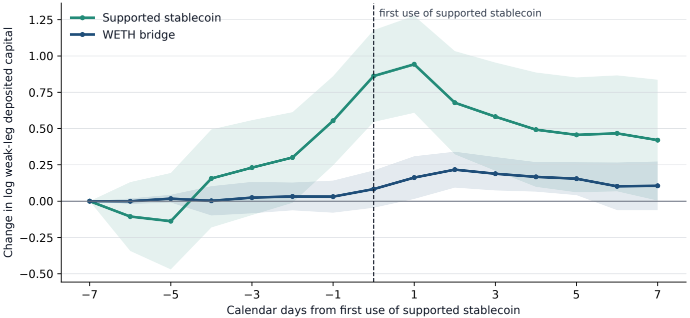
Positive pool capital becomes economically useful as both route legs deepen.

Lower divergence loss may attract LPs; issuers or affiliated market makers may also seed conversion pools.

Note: The sample follows 865 pair events after persistent support appears. The estimates compare first use within 30 days. The supply-side sentence gives two economic interpretations: smaller relative-price movements imply smaller constant-product loss versus holding, and issuer-linked capital may seed convertibility. Provider profitability additionally depends on entry prices, fee income, hedges, and expectations.

Bridge capital builds before first use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



BEFORE FIRST USE

$+0.86$

stablecoin bridge, day -7 to 0

AFTER FIRST USE

-0.44

stablecoin bridge, day 0 to $+7$

Capital builds into first use, then partly unwinds.

Note: 246 pair events with first use 7 to 119 days after persistent support. Relative to WETH, capital changes $+0.78$ before and -0.46 after first use. 92.5% of first-use capital lies in pools active one week earlier, so the rise mainly deepens existing bridges. Capital is lagged one day; intervals cluster by pair and adoption date. First use is endogenous.

Price competition usually changes the venue

SHARE OF ROUTES WITH A QUOTE MORE THAN 1 BP BETTER THAN EXECUTION

1. SAME VEHICLE, USED VENUES → 2. SAME VEHICLE, ALL EXACT VENUES → 3. ANY VEHICLE OR DIRECT

1. SAME VEHICLE, USED VENUES	2. SAME VEHICLE, ALL EXACT VENUES	3. ANY VEHICLE OR DIRECT
6.6% of observed routes	**44.5%** of observed routes	**46.4%** of observed routes

2.0 pp
added by opening other vehicles and the direct path

-1.2 pp
change in stablecoin vehicle share

Note: Routes are repriced for the same pair and input immediately before execution. The sample contains 777,203 routes on 73 monthly dates across exact Uniswap v2, SushiSwap v2, and Uniswap v3 states; minimum input is $100 and the impact limit is 5%. Quotes are gross of gas. The next slide independently quotes feasible stablecoin and WETH paths.

Current prices discipline persistence

AT LEAST 120 DAYS AFTER PAIR ENTRY; BOTH VEHICLE PATHS FEASIBLE

ENTRY VEHICLE GIVES MORE OUTPUT

93.4%

of routes keep the entry vehicle

$\longleftrightarrow$

OTHER VEHICLE GIVES MORE OUTPUT

22.4%

of routes keep the entry vehicle

The first vehicle survives primarily while its path remains price competitive.

Note: For the same pair, input, and pretrade state, we independently quote the best feasible stablecoin and WETH paths. The leader improves gross output by more than 1 bp; every leg remains below 5% own-price impact. The other-vehicle group contains 8,807 routes in 1,311 pairs. Pair-day weighting gives 94.6% when the entry vehicle leads and 24.5% when the other vehicle leads. The comparison is descriptive, gross of gas, and limited to exact venues.

New pairs reshape currency dominance

Network growth carries the rotation

Pair formation and shifts across pairs carry the aggregate change.

Current prices discipline persistence

The first vehicle carries 93.4% of routes while its path leads.

Two-leg depth makes a bridge competitive

Stablecoin use rises with depth relative to WETH.

New pairs choose the bridges shaping dominance.

Appendix map

Definitions

A1 Routes and vehicles
A2 Assets and pegs
A3 Sample exclusions

Validation

A4 Pretrade state
A5 Pool pricing state
A6 Uniswap v1

Robustness

A7 Timing and venue
A8 Count and value
A9 Route endpoints

Markets

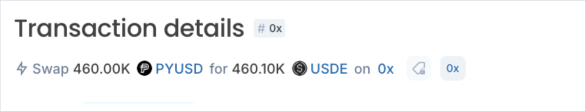
A10 Capital and depth
A11 Coordination
A12–A13 References

Further results

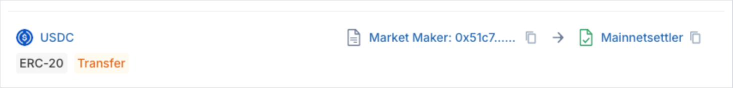
A14 Network position
A15 Endpoint demand
A16–A17b Pair composition
A18 Depth and reach
A19 Pool formation
A20 V4 participation

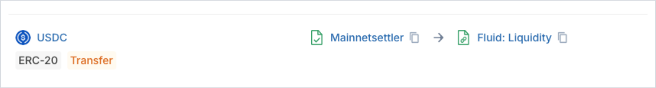
Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER



THE DECISIVE SETTLEMENT TRANSFERS

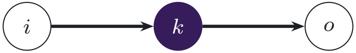




The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD → USDe.

Note: Transaction `0xbda1d07f...9bf67ad9`, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory. Disconnected groups comprise 6.2% of reconstructed components in 2026. Treating each internally valid component as a separate route raises stablecoin count dominance from 42.1% to 43.7%.

A1. One reconstructed route is one unit

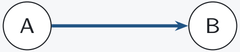


one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

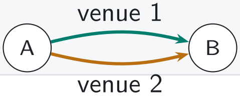
How we reconstruct a route from one transaction

Direct swap
one transaction

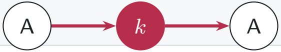


Sequential vehicle route
one transaction

Parallel split
one transaction

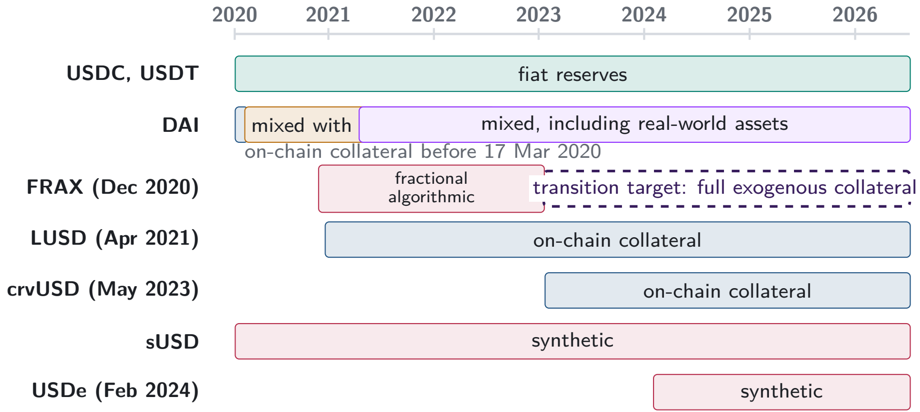


Round trip
one transaction



Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level routes reveal that choice inside aggregate turnover.

A2. Stablecoin backing changes over time



Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

Peg parity preserves currency identity

Bahamian dollar $\longleftrightarrow$ **U.S. dollar**

one-for-one peg

Separate currency, monetary arrangements, and exchange controls

USDC $\longleftrightarrow$ **USDT** $\longleftrightarrow$ **DAI**

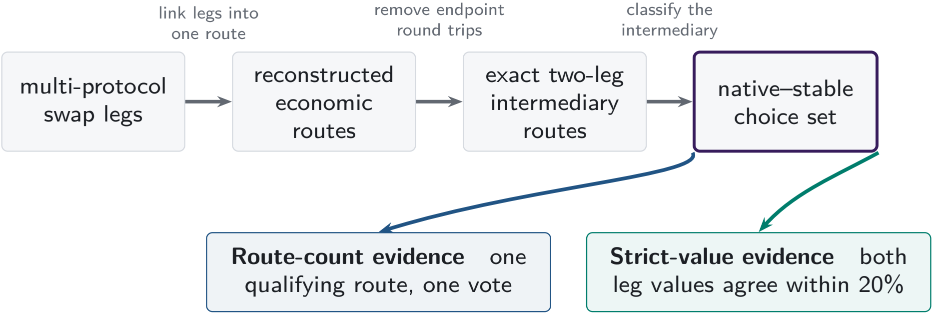
dollar target

Separate issuers, redemption, peg risk, and pool liquidity

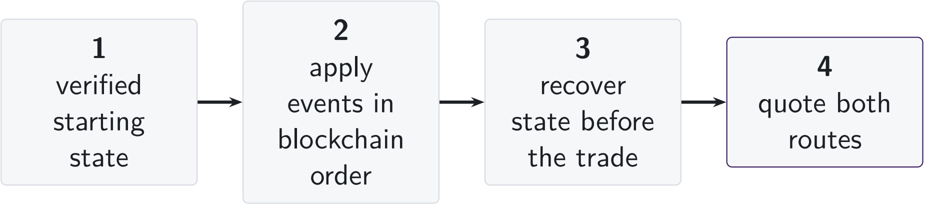
Lower divergence loss may attract liquidity providers; issuers or affiliated market makers may also seed conversion pools. Peg stress reverses the comparison.

Note: The Bahamian-dollar example follows the IMF's 2025 Article IV description of the conventional one-for-one U.S.-dollar peg. The supply-side sentence gives economic interpretations. Separating independent from issuer-linked liquidity and measuring provider returns would require provider identities, funding relationships, entry prices, fees, hedges, and expectations.

A3. One route universe supports two measurements



A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

Executable-quote validation

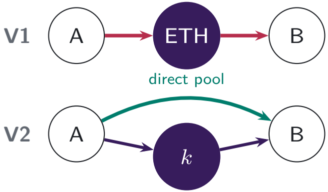
■ recovered pricing input
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\longrightarrow$ V2 MARKET CHOICE



V1 admits only ETH–token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth is path-specific

V3 CONCENTRATED-LIQUIDITY POSITIONS

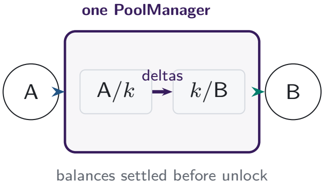
LPs choose a price range and fee tier for each atomic pool. Positions covering the current price determine active liquidity in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



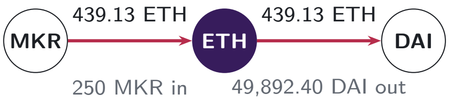
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes where route obligations settle.

A6. V1 forced routes have an on-chain signature

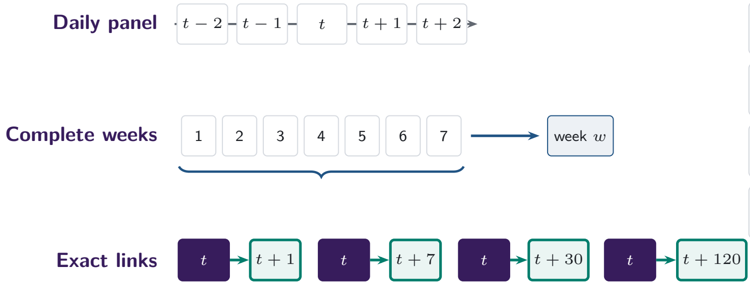
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction `0x4dca160d...96a0ca16` (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

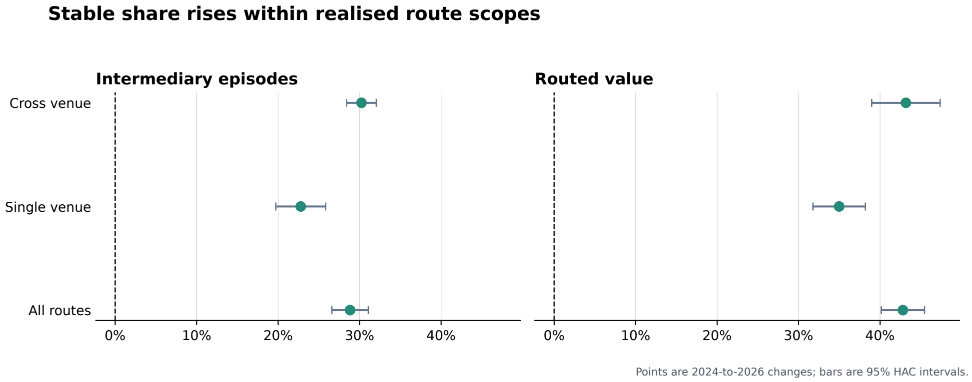
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

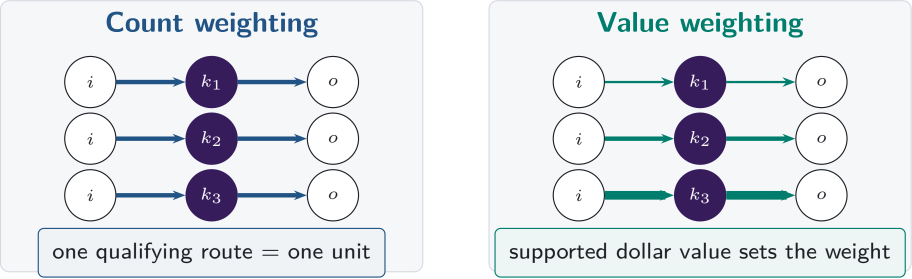
Exact horizons use the required target calendar date.

A7b. Stablecoin share rises within each venue scope



Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

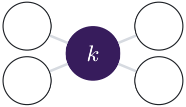
$D_k =$ use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

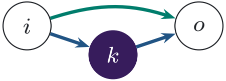
Excess vehicle use

$X_k = D_k \div E_k$
E_k: endpoint share of k

$X_k > 1$: use exceeds the endpoint benchmark



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_j n^I_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$E^N_{k,t} = \frac{n^E_{k,t}}{\sum_j n^E_{j,t}}$$

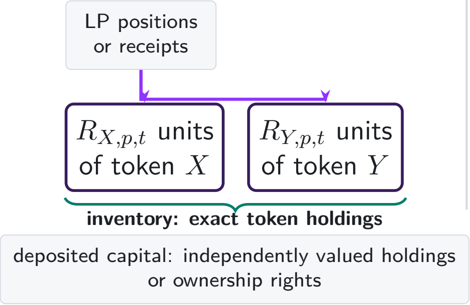
How often k appears at either observed route endpoint.

Observed pool-route endpoints supply the benchmark.

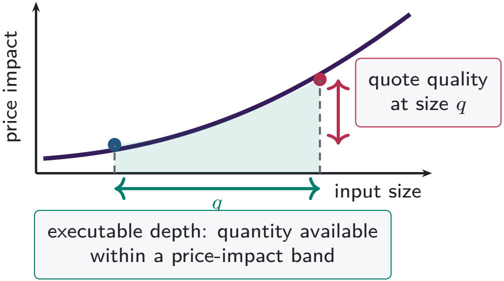
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ

Pool state at time t

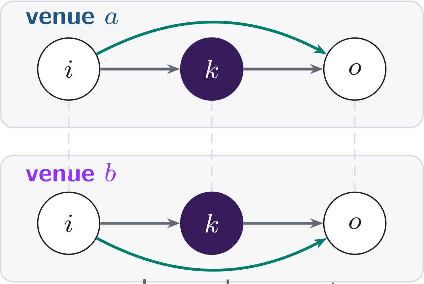


Execution from that state



A11. Additional venues expand the feasible route set

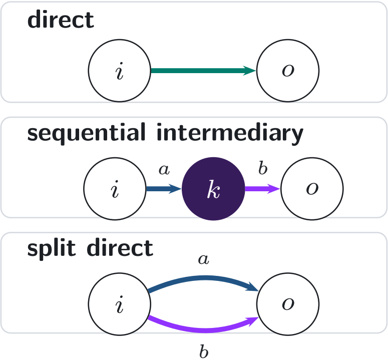
Feasible venue layers



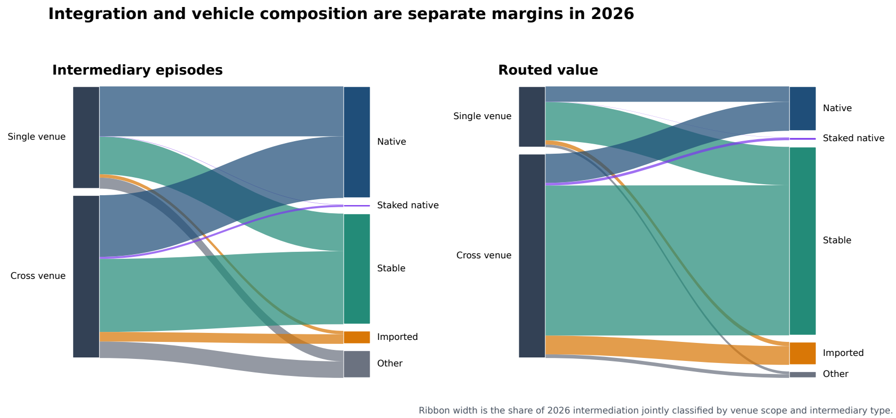
venue layers share assets;
available links define reach

realised
execution

Realised route topology



A11b. Venue scope and vehicle type in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
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- Gopinath and Stein (2021), *Quarterly Journal of Economics*
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- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
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A14. WETH stays the graph hub as realised use shifts

ALL ROUTE LENGTHS; MONTHLY TRADING GRAPHS

WETH BETWEENNESS

#1

in all 8 annual graphs

0.925

in 2026 H1

WETH REALISED SHARE

76.2%

2024

42.4%

2026 H1

USDC + USDT REALISED SHARE

14.9%

2024

37.2%

2026 H1

Binary network position changes much less than actual intermediary use.

Note: Betweenness is approximate and unweighted in annual trading graphs built from trades on the fifteenth day of each month, using 256 source nodes and a fixed seed. Realised shares use all intermediary positions on the same dates. A graph edge records an observed leg, not its depth or execution cost.

A15. Endpoint demand predicts intermediary use

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	$+34.56$	$+34.55$	$+0.05$	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	$+2.42^{*}$	$+2.42^{*}$	$+0.85^{*}$	—
	(1.32)	(1.32)	(0.50)	—
Endpoint-demand share ($D_{a,t}$)	—	—	$+1.59^{***}$	$+0.98^{***}$
	—	—	(0.03)	(0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

After date and currency effects, endpoint demand passes through almost one for one into intermediary use.

Note: 1,828,862 currency-days over 2,259 dates. Parentheses report currency/date-clustered standard errors. The outcome and demand regressor are in pp. Currency effects absorb time-invariant asset-class indicators. Asterisks $*$, $**$, and $***$ denote significance at 10%, 5%, and 1%, respectively.

A16. Pair composition contains a mechanical component

WETH

WETH is an endpoint

↓

Eligible intermediaries: stablecoins

$+21.1$ **pp** **All matched pairs**

WETH-linked route-count mass: 26.6% $\rightarrow$ 46.8%

$+3.7$ **pp** **Pairs without WETH endpoints**

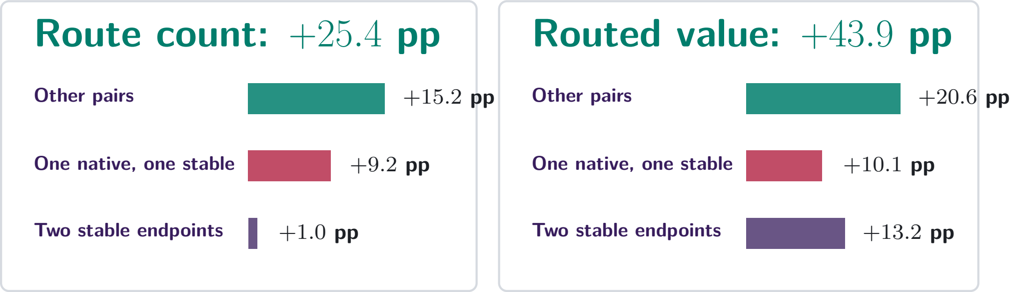
79,347 matched groups

Changes within fixed pairs are $+0.2$ pp overall and $+0.3$ pp after removing WETH endpoints.

Note: WETH cannot be both a pair endpoint and the intermediary. Exact two-leg native-WETH-versus-stablecoin routes on common month-days; fixed-pair estimates also absorb month-day and route scope.

A17. Endpoint direction splits count and value

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE



USDT supplies $+13.7$ pp of the $+13.2$ pp two-stable-endpoint value channel.

Note: Exact additive decomposition on 181 matched dates in 2024 and 2026. Endpoint direction describes the pair; the vehicle remains the intermediary token.

A17b. Rotation survives time and endpoint restrictions

-0.4 **pp to** $+1.0$ **pp**

within-pair term across all adjacent H1 comparisons

1.1% $\rightarrow$ **9.0%**

route-count share with nonvehicle endpoints

0.9% $\rightarrow$ **39.1%**

supported-value share with nonvehicle endpoints

Pair entry and exit contribute $+7.2$ pp by count and $+33.8$ pp by value.

Note: Adjacent comparisons use matched January–June dates from 2019 through 2026. The endpoint restriction removes every pair with WETH or a stablecoin at either endpoint. All changes and components are in pp; supported value applies the 20% agreement rule.

A18. Local depth remains informative alongside network reach

84.1%
of routes use the deepest supported bridge

$+6.79$ **pp**
route share per log point of weak-leg capital

$+9.10$ **pp**
route share per log point of same-day network reach

Local two-leg depth and broader network position both matter.

Note: Five vehicle currencies compared within the same set defined by pair, date, and route scope. Depth is prior-calendar deposited capital on the weaker leg. The regression absorbs set and vehicle effects and clusters by pair and date.

A19. First-use capital lies in active pools

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE

92.5%
of first-use capital lies in continuing pools

$+0.16$
continuing-pool growth relative to WETH

$+6.9$ **pp**
greater probability of a newly active pool

Note: 246 first-use events. A newly active pool has positive prior-calendar capital on day zero and none on day -7; continuing pools are positive on both dates. Difference inference clusters by pair and adoption date.

A20. V4 participation broadens after internal routing

10 PP MORE INTERNAL SAME-ASSET ROUTING

DAYS 1 TO 30

$+0.086^{***}$

actions by previously active origins

DAYS 31 TO 120

$+0.153^{***}$

origins first active after measurement

PERSISTENT VOLATILITY

$+0.318^{***}$

additional later first-active-origin association

Note: V4 vehicle-days with 180 prior days used to classify origins. Specifications absorb vehicle and date effects, include origin-day controls, and cluster by date. Transaction origins proxy for account participation. The estimates are predictive.